■ Vellum Research Studio

System Architecture, Requirements & Comprehensive User Manual for Academic Teams

Author & Lead Architect: Amine Zouaghi	**Target OS:** Windows 10 / 11 (64-bit)
Version: 3.0 Production Release	**License:** 100% Free Open-Access (MIT)

1. Executive Summary & Edition Overview

Vellum is an integrated academic research platform combining heavy offline bibliometrics and high-volume full-text PDF harvesting with modern force-directed citation networks, active machine-learning screening, and systematic evidence synthesis. The suite is distributed as two ready-to-use applications sharing one local-first research workflow:

- **Vellum Desktop V2.0 — Fernly Botanical Bento (Port 8501):** Native Python/Streamlit workbench capable of parallel downloading 10,000+ DOIs, deduplicating Scopus/WoS/PubMed exports, active-learning screening, citation-force-graph exploration, and local PDF vault management.
- **Vellum Web — Fernly Botanical Bento (Port 3002):** Flagship web workspace featuring the Connected Papers interactive network canvas, ASReview active-learning screening, visual clause-by-clause boolean search, batch multi-tier PDF downloader, and tri-pane systematic notebooks.

2. System & Hardware Requirements

Category	Specification	Notes & Recommendations
Operating System	Windows 10 / 11 (64-bit)	Windows 10 v1903 or newer recommended
RAM	4 GB min / 8 GB rec	8 GB optimal for 5,000+ candidate screening datasets
Desktop Engine	Zero software setup required	Standalone binary included (Vellum.exe). No Python install needed.
Web Editions	Zero setup (one-click package) or Node.js 18+ (from source)	The shareable package Vellum_Web_Fernly_SHARE.zip bundles its own portable Node.js runtime, verified against nodejs.org checksums. Running from source still needs Node.js LTS 20+.
Browser	Chromium / Firefox / Edge	Hardware-accelerated HTML5 canvas enabled for graph physics
Network	Outbound HTTPS (Port 443)	Connects to OpenAlex, arXiv, Crossref, and Semantic Scholar APIs

3. 1-Click Application Launchers

Each application lives in its own self-contained folder and can be launched directly without command-line configuration:

File to Double-Click	Target Application	Local URL & Action
`Vellum_Desktop\Vellum-Desktop-V2.0-Setup-x64.exe`	Vellum Desktop Installer	Installs per-user, no admin rights • Start Menu + uninstaller
`Vellum_Web\Vellum_Web_Fernly_SHARE\START_VELLUM_WEB.bat`	Vellum Web — Fernly Bento (one-click package)	The package to send to a colleague • **nothing to install** • Node.js bundled

`Vellum_Web\2_launch_fernly_web_app.bat`	Vellum Web — Fernly Bento (Flagship)	http://localhost:3002 • Force Graph, AI Screening, Boolean Search
`Vellum_Desktop\1_launch_desktop_app.bat`	Vellum Desktop V2.0 Workbench	http://localhost:8501 • High-volume batch PDF extraction
`3_launch_both_apps.bat`	Dual Suite Concurrent Launch	Boots Desktop (8501) and Web (3002) simultaneously

First run of the installer: it is not code-signed, so Windows SmartScreen shows “Windows protected your PC”. Choose **More info** → **Run anyway**. To confirm the file you received is the one that was built, run certutil -hashfile Vellum-Desktop-V2.0-Setup-x64.exe SHA256 and compare it with DISTRIBUTABLES\SHA256SUMS.txt.

4. Key Features & Operational Workflows

■■ **Connected Papers Interactive Citation Network:** Provides a 3-panel exploration engine modeled after Connected Papers. Nodes are sized by lifetime citations and colored via a continuous mint-to-pine publication year gradient. Features origin paper halo rings, drag-and-drop Coulomb-Hooke physics, floating navigational guide, and direct tabs for **Prior Works** (foundational ancestry) and **Derivative Works** (forward breakthroughs). Any node can be re-centered with *'Build graph from this paper'*.

■ **Desktop-Style Clause-by-Clause Visual Boolean Builder:** Located under Academic Search. Allows researchers to build exact queries with an initial WHERE clause followed by AND, OR, and NOT connectors. Field dropdowns automatically map terms to target index tokens ([Title/Abstract], au:, year:, topic:) with real-time compilation and 1-click execution.

■ **ASReview Active-Learning Screening Hub:** Enables machine-learning assisted title and abstract screening. Accepts multi-file uploads (.csv, .ris, .bib) with automated deduplication across overlapping sources. Employs TF-IDF feature extraction with Naive Bayes probability ranking, keyboard hotkeys ([I] Include, [E] Exclude, [U] Uncertain), session saving/switching, and live recall stopping curves.

■ **Multi-Tier Failover PDF Harvesting Engine:** Integrated into both web and desktop versions. Automatically searches across Unpaywall, publisher Open Access APIs, Semantic Scholar, and academic mirrors while bypassing publisher paywall traps. Delivers binary PDFs directly into local browser memory and downloads folder.

■ **Grounded Tri-Pane Evidence Synthesis Workspace:** A local-first systematic review notebook workspace. 100% private and persistent using browser IndexedDB. Left pane stores sources and PDFs; center pane provides an interactive document reader; right pane serves as an evidence synthesis studio for recording verbatim claims and generating citation exports.

5. Troubleshooting & Technical Guidance

- **Port Occupied (EADDRINUSE):** If port 3002 is already in use, open PowerShell and terminate previous instances using Get-Process -Name node | Stop-Process -Force, then re-run the launcher.
- **Data Privacy & Storage:** Web notebooks and active screening sessions are stored locally in IndexedDB (Dexie.js). No account, API key, or cloud transmission of personal data is required.
- **API Rate Limits:** Click the **'API Tokens'** button in the Academic Search hub to supply free personal tokens for OpenAlex, Semantic Scholar, or Crossref for unrestricted high-volume querying.